Section 7: Regression

Section 7 Overview

1. **Regression equations**
2. Correlation vs. Causation

Regressions

What does a regression of one variable on another variable tell us?

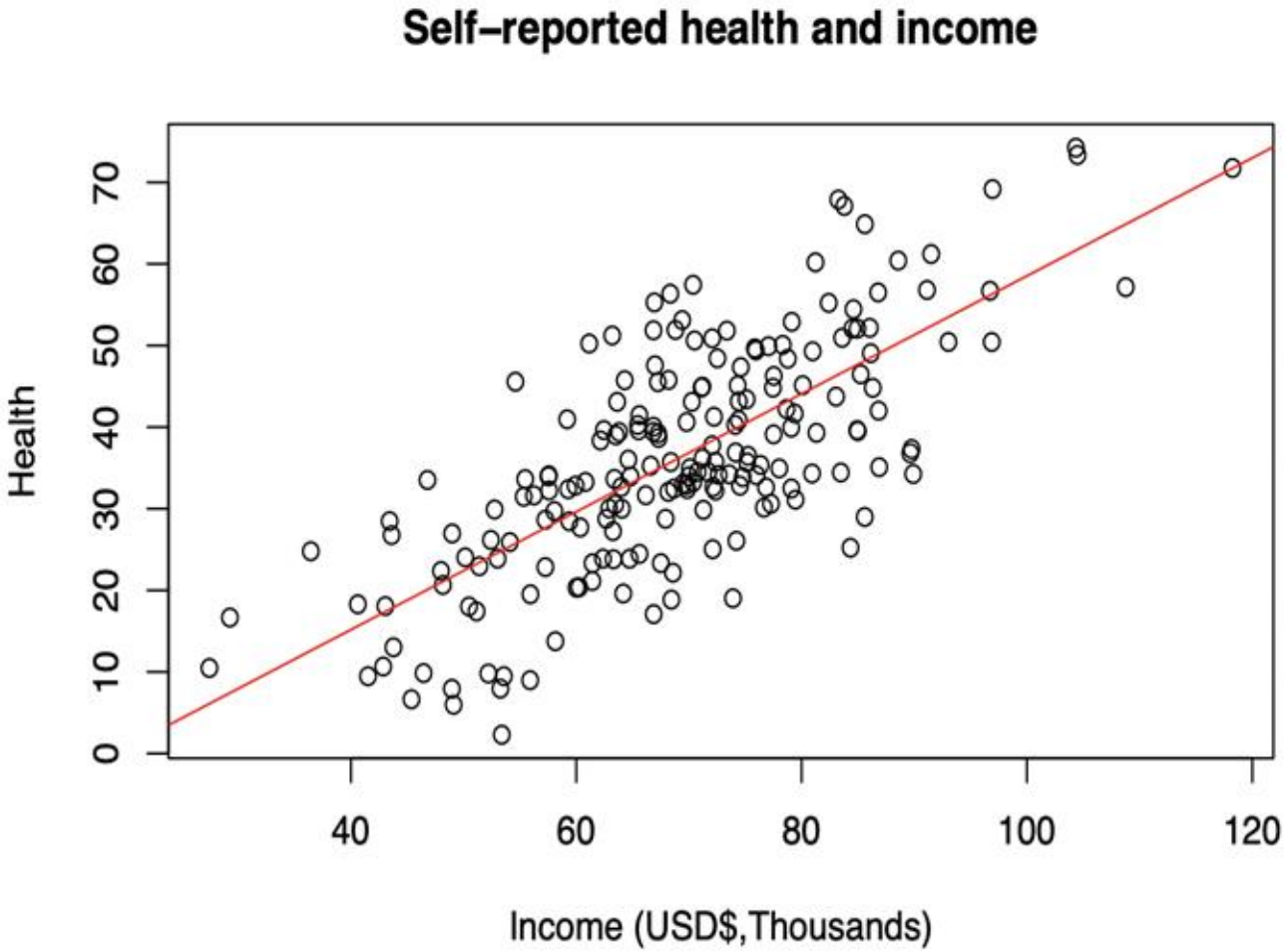
Why might we want to use regressions?

What is an example of when a regression might be useful?

What do regressions not tell us?

Health and Income

How are these variables related? How strong is the relationship?



A Simple Linear (Bivariate) Regression

General form: Y = a + b*X + e

- a: y-intercept
- b: slope
- e: error

Why does this error term exist?

In the health vs. income relationship, what are some factors that could be in the error term?

Bivariate Regression – Health vs. Income

What is the equation of the regression line?

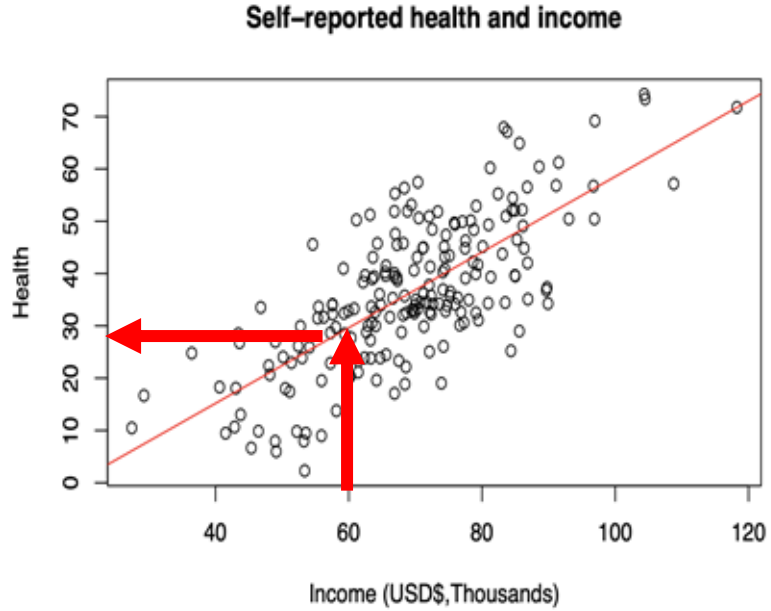
How do we state the coefficient on income in words?

How do we know whether the relationship is statistically significant?

How could we predict the health of somebody of a given income? (e.g. $60K)

Table 1:

	Dependent variable:
	Health
Income (thousands)	0.723^{***} (0.630, 0.816)
Constant	-13.735^{***} $(-20.305, -7.164)$
Observations	200
Note:	*p<0.1; **p<0.05; ***p<0.01



Adding Controls (Multivariable Regression)

General form: $Y = a + b1*X1 + b2*X2 + \ldots + e$

- a: y-intercept
- b1, b2, …: slopes
- e: error

Why would we want to include other control variables in the model of health vs. income?

What other control variables may we want to add?

Multivariable Regression – Health vs. Income

We now add “age” as an additional control :

Does the sign of the coefficient on “age” make sense?

How do we interpret the coefficient on income?

Does the coefficient on income remain statistically significant?

Does the coefficient on income change? What does this mean?

Table 2:

	Dependent variable:
	Health
Income (thousands)	0.731^{***} (0.689, 0.773)
Age	-0.548^{***} (-0.586, -0.510)
Constant	10.657^{***} (7.258, 14.055)
Observations	200
Note:	*p<0.1; **p<0.05; ***p<0.01

Should we just control for everything?

We now add “cats” as an additional control:

Does the size of the coefficient on “cats” make sense?

Does the coefficient on income remain statistically significant?

You, the discerning consumer of research, should determine whether what you are reading makes sense, given the regression specification! There is not always a clear-cut answer.

Table 4:

	Dependent variable:
	Health
Income (thousands)	0.743^{***} (0.716, 0.770)
Age	-0.548^{***} (-0.573, -0.523)
Race White	6.474^{***} (5.694, 7.254)
Number of cats	0.002 (-0.077, 0.080)
Constant	6.238^{***} (3.850, 8.625)
Observations	200
Note:	*p<0.1; **p<0.05; ***p<0.01

A follow-up example

A new (hypothetical) paper regresses "survival rate of craniotomy" on "size of teaching hospitals," and finds that the larger teaching hospitals have worse survival rates. What can we conclude?

A reviewer tells the authors to control for underlying patient risk. The authors re-run the regression (with the new control), and find that larger teaching hospitals are now associated with better survival rates. What can we conclude?

Moving Toward Causality

Have we established a causal relationship between income and health?

What would we have to do to establish a causal relationship between income and health?

Direction of Causality

What if we were able to control for every variable that might influence both health and income?

In that case, would we know for sure that income causes an increase in health?

What if better health leads to higher income? Do our regression results tell us about which causes which?

- **Can you come up with an explanation for why the relationship might run in this direction (from health → income)?**

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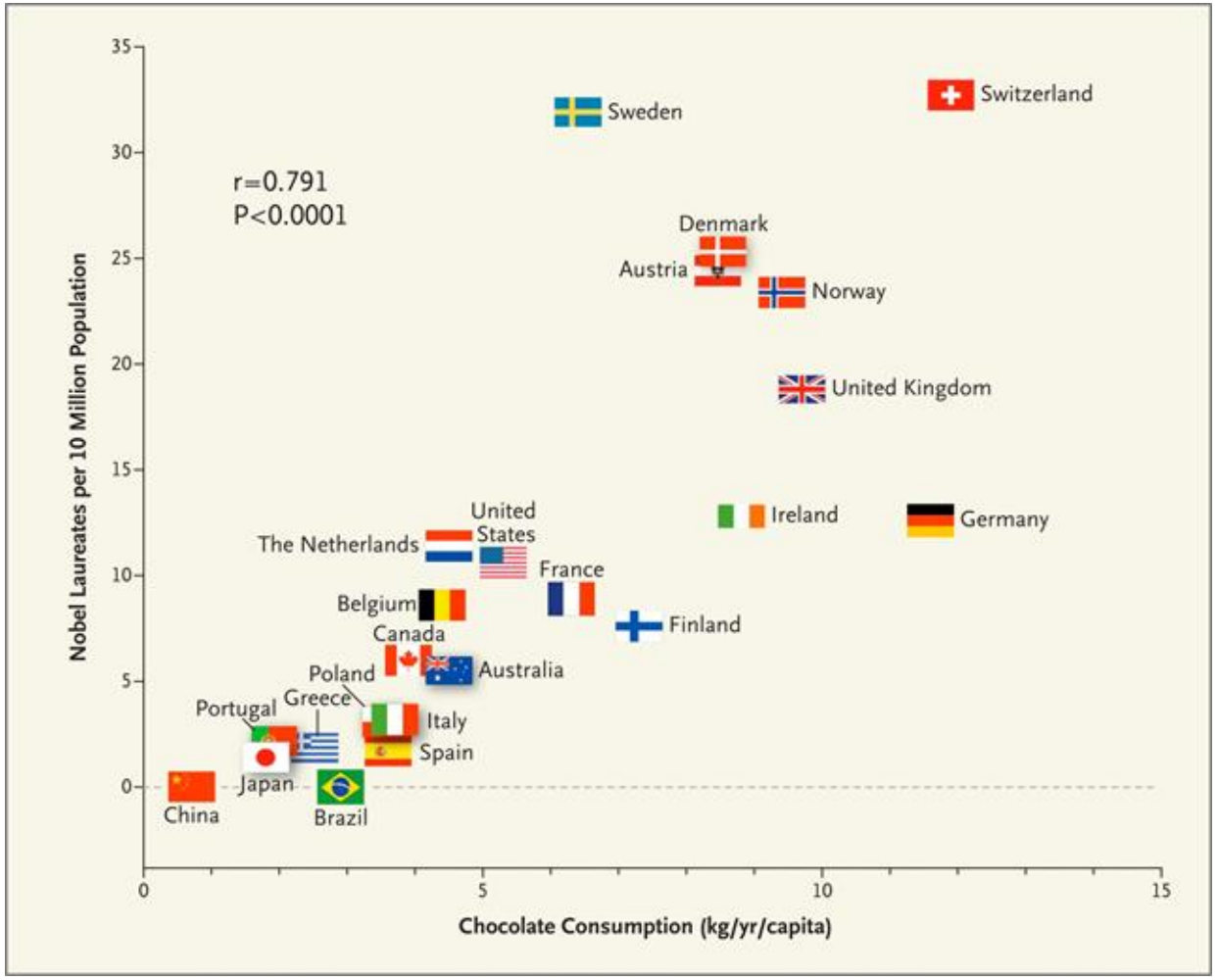
1. Regression equations
2. **Correlation vs. Causation**

Correlation vs. Causation

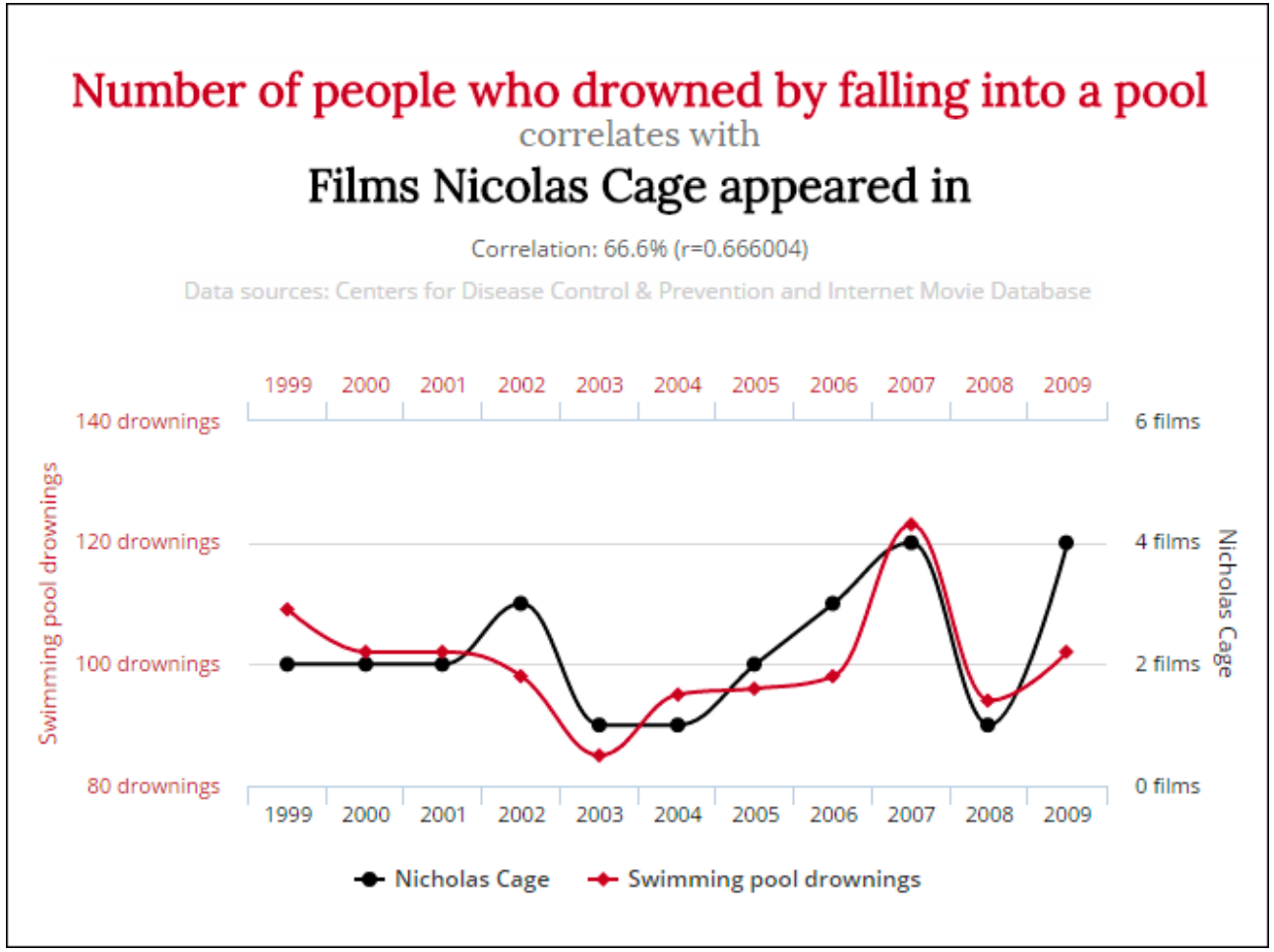
Reasons we may see a correlation, but it may not be causation:

- Third variable that explains/causes both (confounder)
- Random coincidence
- Reverse causality (causal relationship exists, but we got it backwards)

Correlation vs. Causation Examples



Correlation vs. Causation Examples



Correlation vs. Causation Examples

Medicaid Is Worse Than No Coverage at All

New research shows that patients on this government plan fare poorly. So why does the president want to shove one in four Americans into it?

- Evaluate this claim from an Op-Ed in the Wall Street Journal.

- Basic gist of the papers the Op-Ed author cites:
  - Patients w/ Medicaid were more likely to die / get sicker than privately insured (or uninsured in some cases)
  - Many studies controlled for socioeconomic status

• *Head and neck cancer*: A 2010 study of 1,231 patients with cancer of the throat, published in the medical journal Cancer, found that Medicaid patients and people lacking any health insurance were both 50% more likely to die when compared with privately insured patients —even after adjusting for factors that influence cancer outcomes. Medicaid patients were 80% more likely than those with private insurance to have tumors that spread to at least one lymph node. Recent studies show similar outcomes for breast and colon cancer.

• *Major surgical procedures*: A 2010 study of 893,658 major surgical operations performed between 2003 to 2007, published in the Annals of Surgery, found that being on Medicaid was associated with the longest length of stay, the most total hospital costs, and the highest risk of death. Medicaid patients were almost twice as likely to die in the hospital than those with private insurance. By comparison, uninsured patients were about 25% less likely than those with Medicaid to have an "in-hospital death." Another recent study found similar outcomes for Medicaid patients undergoing trauma surgery.

• *Poor outcomes after heart procedures*: A 2011 study of 13,573 patients, published in the American Journal of Cardiology, found that people with Medicaid who underwent coronary angioplasty (a procedure to open clogged heart arteries) were 59% more likely to have "major adverse cardiac events," such as strokes and heart attacks, compared with privately insured patients. Medicaid patients were also more than twice as likely to have a major, subsequent heart attack after angioplasty as were patients who didn't have any health insurance at all.

Correlation vs. Causation Examples



Medicaid Is Worse Than No Coverage at All

New research shows that patients on this government plan fare poorly. So why does the president want to shove one in four Americans into it?

- Are you convinced that Medicaid causes poor health or bad outcomes?

- What are some factors (confounders) that might explain this relationship?

Questions?